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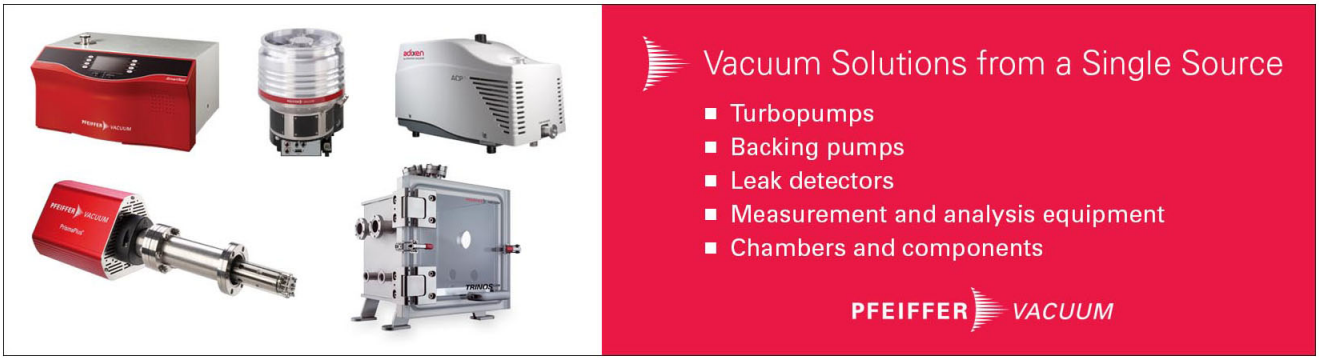

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Evaluation of $1/\nu$ neoclassical transport in stellarators

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Using an analytic solution of the kinetic equation in the $1/\nu$ regime, a new formula for the neoclassical transport coefficients is obtained which takes into account all classes of trapped particles. This formula holds in any coordinate system and no simplifying assumptions about the magnetic field are needed. Therefore it is also applicable to complex magnetic fields given in real space coordinates. The method and the results can be used to optimize magnetic field configurations with respect to the $1/\nu$ regime. The method is bench-marked against Monte Carlo calculation both for the $l=3$ classical stellarator model and also for the original Helias configuration [J. Nührenberg and R. Zille, Phys. Lett. A **114**, 129 (1986)] with a more complex magnetic field structure. Some features of transport for Helias are clarified by analyzing the bounce-averaged drift velocity. © *1999 American Institute of Physics.* [S1070-664X(99)00412-7]

I. INTRODUCTION

A characteristic feature of stellarator magnetic field configurations is the lack of toroidal symmetry and, thus, in contrast to tokamaks, a full three-dimensional description is necessary. The absence of magnetic field symmetry leads to an enhanced drift of locally trapped particles (banana drift) across the magnetic surface and, therefore, to enhanced neoclassical losses in the low-collisional regime. This effect has been studied theoretically[1–5] as well as numerically (see, e.g., Refs. 6 and 7). In general, various transport regimes may be realized in stellarator devices. At the same time, of particular interest for the optimization problems of stellarators is the so called $1/\nu$ regime. In this regime the neoclassical transport coefficients increase with decreasing collision frequency. This regime is directly caused by the stellarator magnetic field asymmetry. Minimization of neoclassical transport in the $1/\nu$ regime is one of the key issues in stellarator optimization.[8]

At present, neoclassical transport coefficients are often computed using kinetic equation solvers (see, e.g., Ref. 9) which utilize the magnetic coordinate system. In optimization problems, many iterations of this computation are needed and therefore a faster and simpler method is desirable, especially, in the case when the Fourier spectrum of the magnetic field in magnetic coordinates is broad. If such a method works both in flux coordinates and in real space coordinates, then, starting from the coil system, one has the choice to use either flux coordinates generated by a chain of appropriate codes (see, e.g., Ref. 10) or to use the magnetic field directly if these codes are not available or finite β effects are not of interest.

In the present paper a new numerical method of calculating the neoclassical transport coefficients is proposed for the $1/\nu$ transport regime using an analytic solution of the banana kinetic equation. In standard neoclassical theory (see, e.g., Refs. 2 and 5) such a solution is obtained using simplified models of the magnetic field and taking into account only particles trapped within a single ripple well. Thus, the contributions from other classes of trapped particles which pass during their bounce period two or more magnetic field ripples are neglected. In complex magnetic fields with, e.g., additional toroidal ripple modulation, such an approximation will lead to an underestimate of the $1/\nu$ transport coefficients. As in Ref. 11, in the present paper the solution to the kinetic equation is obtained without any simplifying assumptions about the magnetic field and, at the same time, all classes of trapped particles are taken into account. The resulting expressions for the neoclassical transport coefficients are identical to the standard neoclassical transport coefficients where the ripple modulation amplitude ϵ_h is replaced by the effective ripple modulation amplitude ϵ_{eff}, which describes the effect of the magnetic field geometry upon $1/\nu$ transport. The amplitude ϵ_{eff} is expressed in terms of a weighted integral of the geodesic curvature along the magnetic field line covering the magnetic surface and can be easily evaluated numerically. The present approach is closely related to the method of computing transport coefficients in the Pfirsch–Schlüter regime developed in Ref. 12. The derivation of the formula for the effective ripple amplitude ϵ_{eff} is given in Sec. II A. In Sec. II B it is shown that the formula gives zero $1/\nu$ transport for quasi-helically symmetric configurations.

The proposed method has been applied to two Helias-type configurations[13,14] and to the Uragan-3M torsatron[15] (U-3M). These configurations are described in Sec. III. The results of the numerical evaluation of ϵ_{eff} for these configurations as well as some details of the numerical procedure are presented in Sec. V A. Some of the results obtained are then compared to Monte Carlo calculations in Sec. V B. In order to clarify some features of the results for the

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Helias configurations, the drift motion of trapped particles across the magnetic surface is analyzed in Sec. IV with the help of the technique considered in Refs. 11 and 16.

II. SEMI-ANALYTICAL FORMULA FOR $1/\nu$ TRANSPORT COEFFICIENTS ALLOWING FOR PARTICLES TRAPPED WITHIN MORE THAN ONE RIPPLE WELL

A. Formula derivation

In the following we use spatial coordinates $\mathbf{x}=(x^1,x^2,x^3)$ which satisfy $\mathbf{h}\cdot\nabla x^{1,2}=0$ where $\mathbf{h}=\mathbf{B}/B$ is the unit vector along the magnetic field. These coordinates $x^1=\psi$, $x^2=\theta_0$, $x^3=\zeta$ are the magnetic surface label, the poloidal angle-like variable and the toroidal angle, respectively. Thus, a single field line is described by $x^1=$const, $x^2=$const and x^3 varying in the infinite limits. The independent variables in velocity space are the total energy $w=mv^2/2+e\phi$ and the perpendicular adiabatic invariant $J_\perp=v_\perp^2/B$, where m and e are the particle mass and charge, and ϕ is the electrostatic potential. The parallel velocity can then be expressed as

$$v_\parallel(\mathbf{x},w,J_\perp)=\sigma\sqrt{\frac{2}{m}(w-e\phi)-J_\perp B},\quad (\sigma=\pm 1). \tag{1}$$

For the description of Coulomb collisions the Lorentz model is used,

$$\mathrm{St}_L(f)=\frac{\nu A}{\sin\chi}\frac{\partial}{\partial\chi}\left(\sin\chi\frac{\partial f}{\partial\chi}\right)=\frac{4\nu A v_\parallel}{B}\frac{\partial}{\partial J_\perp}\left(J_\perp v_\parallel\frac{\partial f}{\partial J_\perp}\right), \tag{2}$$

where χ is the pitch-angle, and $\nu A(v^2)$ is the pitch-angle scattering frequency.

Let us consider a single magnetic field line fixed by the values of x^1 and x^2. Then the local minima of the magnetic field module B within an interval of arbitrary size, $0<x^3<L_\zeta$ can be numbered with index k, where $1\leqslant k\leqslant M$ and M is a total number of minima in the given interval. With respect to these minima, the trapped particles can be defined as those whose motions are restricted to the intervals $0<x^3_{\min,k}<x^3<x^3_{\max,k}<L_\zeta$, where the reflection points satisfy $v_\parallel^2(x^3_{\min,k})=v_\parallel^2(x^3_{\max,k})=0$ and $v_\parallel^2(x^3)>0$. Here $v_\parallel$ is given by (1) with fixed $x^{1,2}$, w and $J_\perp$. With such a definition, the trapped particle index k is unique only for the class of particles trapped within a single ripple well. The particles, which pass two or more minima during one bounce oscillation, are labeled by two or more values of the index k, respectively. Classes of differently trapped particles are separated by the boundary surfaces,

$$J_\perp=J_{\perp b}^{(k,l)}(x^1,x^2,w),\quad J_{\perp b}^{(k,l)}\equiv\frac{2(w-e\phi)}{mB_{\max}^{(k,l)}},\quad l=1,2,\ldots, \tag{3}$$

where l labels the classes of particles. In each minimum such a classification is different. Each class is connected to a particular maximum, $B_{\max}^{(k,l)}$, which defines the lower boundary of $J_\perp$ allowed for the class l as shown in Fig. 1. Here we assume that the electrostatic potential ϕ is constant on the field line. The number of boundaries and therefore also the number of classes of particles goes to infinity together with L_ζ. (We consider only typical minima which are not close to the ends of the interval.) In the limit $l\to\infty$ the corresponding maxima of B are approaching the maximum value of the magnetic field module on a given magnetic surface $B_{\max}^{(\mathrm{abs})}$. At the same time, the phase space volume occupied by particles which have only one reflection point within the interval $0<x^3<L_\zeta$ becomes small compared to the volume occupied by trapped particles. Therefore such particles are neglected in the consideration below.

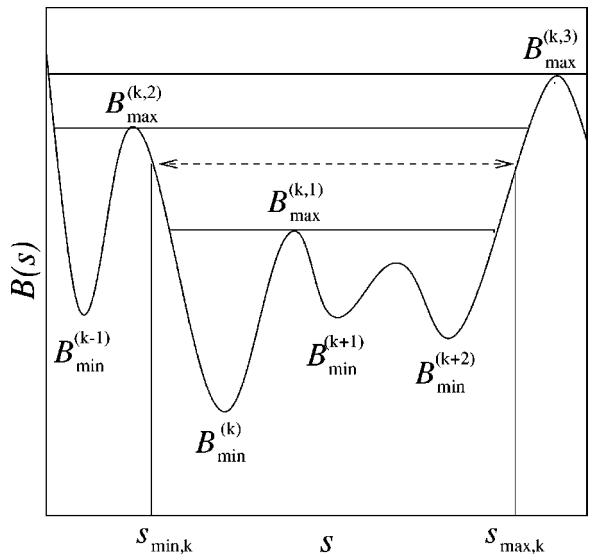

FIG. 1. The magnetic field module as a function of the distance along the field line s. The dashed line is the value of $v^2/J_\perp$ for a particle trapped within 3 local magnetic wells. The turning points $s_{\min,k}$ and $s_{\max,k}$ of this particle are shown. The horizontal solid lines correspond to the boundary values of $v^2/J_\perp=v^2/J_{\perp b}^{(k,l)}$ separating different classes of particles associated with the ripple well k.

The banana (bounce-averaged) kinetic equation for the distribution function of the particles trapped in minimum k corresponding to the Lorentz collision model (2) is

$$\tau_b^{(k)}\frac{\partial f^{(k)}}{\partial t}+\delta x_{(k)}^i\frac{\partial f^{(k)}}{\partial x^i}=4\nu A\frac{\partial}{\partial J_\perp}\left(J_\perp I^{(k)}\frac{\partial f^{(k)}}{\partial J_\perp}\right), \tag{4}$$

where

$$\tau_b^{(k)}=2\int_{x^3_{\min,k}}^{x^3_{\max,k}}\frac{\mathrm{d}x^3}{|v_\parallel|}\sqrt{g_{33}}=2\int_{s_{\min,k}}^{s_{\max,k}}\frac{\mathrm{d}s}{|v_\parallel|}=\oint_{(k)}\frac{\mathrm{d}s}{v_\parallel} \tag{5}$$

is the bounce period,

$$\delta x_{(k)}^i=2\int_{x^3_{\min,k}}^{x^3_{\max,k}}\frac{\mathrm{d}x^3}{|v_\parallel|}\sqrt{g_{33}}\,\mathbf{V}\cdot\nabla x^i$$
$$=2\int_{s_{\min,k}}^{s_{\max,k}}\frac{\mathrm{d}s}{|v_\parallel|}\mathbf{V}\cdot\nabla x^i=\oint_{(k)}\frac{\mathrm{d}s}{v_\parallel}\mathbf{V}\cdot\nabla x^i \tag{6}$$

is the particle cross-field displacement due to the magnetic and electric drift after one bounce period (here $i=1,2$), and

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$$I^{(k)}=2\int_{x^3_{\min,k}}^{x^3_{\max,k}}\frac{\mathrm{d}x^3}{B}|v_\parallel|\sqrt{g_{33}}$$
$$=2\int_{s_{\min,k}}^{s_{\max,k}}\frac{\mathrm{d}s}{B}|v_\parallel|=\oint_{(k)}\frac{\mathrm{d}s}{B}v_\parallel. \tag{7}$$

Here $\mathbf{V}$ is the particle drift velocity and g_{33} is the parallel component of the metric tensor. In Eqs. (5), (6) and (7) different versions are given with x^3 and s, the arc-length measured along the field line, as integration variables.

Let us consider the boundary surface associated with the local magnetic field maximum $B^{(k,l)}_{\max}$ situated between the minima k and $k+1$ (see Fig. 1). Introducing the subscripts $+$ and $-$ for the classes of trapped particles with $J_\perp<J^{(k,l)}_{\perp b}$ and $J_\perp>J^{(k,l)}_{\perp b}$, respectively, one can express the continuity of the distribution function, as

$$f^{(k)}_-=f^{(k)}_+=f^{(k+1)}_-=f^{(k+1)}_+. \tag{8}$$

Using (8) and the conservation of particles in the boundary layer, we obtain the condition for the normal derivatives,

$$\left[I^{(k)}\frac{\partial f^{(k)}}{\partial J_\perp}\right]_+=\left[I^{(k)}\frac{\partial f^{(k)}}{\partial J_\perp}+I^{(k+1)}\frac{\partial f^{(k+1)}}{\partial J_\perp}\right]_-. \tag{9}$$

Here it is used that $f^{(k)}_+=f^{(k+1)}_+$ due to multiple labeling [the same is true for the quantities given by Eqs. (5), (6) and (7)]. At the bottom of a local magnetic well defined by

$$J_\perp=J^{(k)}_{\perp\max}(x^1,x^2,w),\quad J^{(k)}_{\perp\max}\equiv\frac{2(w-e\phi)}{mB^{(k)}_{\min}}, \tag{10}$$

where $B^{(k)}_{\min}$ is the minimum value of the magnetic field in the magnetic well number k, the boundary condition is

$$I^{(k)}\frac{\partial f^{(k)}}{\partial J_\perp}=0. \tag{11}$$

Note that the equation for the passing particles is obtained as a particular case of (4) with $|x^3_{\max,k}-x^3_{\min,k}|\to\infty$. Due to the assumption of small collision frequency compared to bounce frequency used for bounce averaging, the distribution function is independent from x^3 in the region $x^3_{\min,k}<x^3<x^3_{\max,k}$. In this case the distribution function also becomes independent from x^2, because it is constant along the whole magnetic field line which covers ergodically the irrational magnetic surface.

In the following we use the equivalence between the field line averages and the averages over the volume between two neighboring magnetic surfaces notated as $\langle\ldots\rangle$,

$$\frac{\langle P\rangle}{\langle Q\rangle}\equiv\int_0^{2\pi}\mathrm{d}x^2\int_0^{2\pi}\mathrm{d}x^3\sqrt{g}\,P\left(\int_0^{2\pi}\mathrm{d}x^2\int_0^{2\pi}\mathrm{d}x^3\sqrt{g}\,Q\right)^{-1}$$
$$=\oint_\psi\frac{\mathrm{d}S}{|\nabla\psi|}P\left(\oint_\psi\frac{\mathrm{d}S}{|\nabla\psi|}Q\right)^{-1}$$
$$=\lim_{L_s\to\infty}\int_0^{L_s}\frac{\mathrm{d}s}{B}P\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}Q\right)^{-1}, \tag{12}$$

where g is the metric determinant, $\mathrm{d}S$ is the element of the magnetic surface area and the integration in the last expression is along the magnetic field line.

Dividing (4) by

$$2\int_{x^3_{\min,k}}^{x^3_{\max,k}}\frac{\mathrm{d}x^3}{B}\sqrt{g_{33}}=2\int_{s_{\min,k}}^{s_{\max,k}}\frac{\mathrm{d}s}{B},$$

and putting $s_{\min,k}=0$ and $s_{\max,k}=L_s$, with the help of (12), we obtain in the limit $L_s\to\infty$,

$$\left\langle\frac{B}{|v_\parallel|}\right\rangle\frac{\partial f^{(k)}}{\partial t}=4\nu A\frac{\partial}{\partial J_\perp}\left(J_\perp\langle|v_\parallel|\rangle\frac{\partial f^{(k)}}{\partial J_\perp}\right). \tag{13}$$

Note that the term with the derivative over $x^1\equiv\psi$ is absent because of Liouville's theorem, $\nabla\cdot[\mathbf{V}(\mathbf{r},J_\perp,w)B/v_\parallel]=0$. The remaining boundary condition for passing particles at $J_\perp=0$ is

$$\left(J_\perp\frac{\partial f^{(k)}}{\partial J_\perp}\right)_{J_\perp\to0}\to0. \tag{14}$$

Looking for the transport coefficients in the $1/\nu$ regime, we consider the stationary problem, and look for a solution in the form $f^{(k)}=f_0+f^{(k)}_1$ where $f_0=f_0(\psi,w)$ is the Maxwellian distribution. The banana kinetic equation (4) then takes the form

$$\delta\psi^{(k)}\frac{\partial f_0}{\partial\psi}=4\nu A\frac{\partial}{\partial J_\perp}\left(J_\perp I^{(k)}\frac{\partial f^{(k)}_1}{\partial J_\perp}\right), \tag{15}$$

where $\delta\psi^{(k)}\equiv\delta x^1_{(k)}$. The boundary conditions for $f^{(k)}_1$ are given by (8), (9) and (11) with a change of $f^{(k)}$ to $f^{(k)}_1$. The solution of the equation for passing particles (13) in the stationary case with the account of (14) gives $f^{(k)}_1=0$.

Using the explicit expression for the guiding center drift velocity and the fact that for the equilibrium magnetic field $(\nabla\times\mathbf{B})\cdot\nabla\psi=0$, we obtain

$$\frac{\mathbf{V}\cdot\nabla\psi}{|v_\parallel|}=\left(\frac{v^2}{|v_\parallel|}+|v_\parallel|\right)\frac{|\nabla\psi|k_G}{2\omega_c}$$
$$=-\frac{\partial}{\partial J_\perp}\left(v^2|v_\parallel|+\frac{1}{3}|v_\parallel|^3\right)\frac{|\nabla\psi|k_G}{B\omega_c}, \tag{16}$$

where $\omega_c=eB/(mc)$ is the cyclotron frequency and

$$k_G=\mathbf{n}_\psi\cdot(\mathbf{h}\times(\mathbf{h}\cdot\nabla)\mathbf{h}),\quad \mathbf{n}_\psi=\frac{\nabla\psi}{|\nabla\psi|} \tag{17}$$

are the geodesic curvature of the magnetic field line and the unit vector normal to the magnetic surface, respectively. Substituting (16) in (6), we obtain for $\delta\psi^{(k)}$,

$$\delta\psi^{(k)}=-\frac{1}{3}\frac{\partial H^{(k)}}{\partial J_\perp}, \tag{18}$$

$$H^{(k)}=2\int_{s_{\min,k}}^{s_{\max,k}}\mathrm{d}s|v_\parallel|(3v^2+v_\parallel^2)\frac{|\nabla\psi|k_G}{B\omega_c}. \tag{19}$$

Thus, from (15) we obtain for the derivative of $f^{(k)}_1$,

$$\frac{\partial f^{(k)}_1}{\partial J_\perp}=-\frac{H^{(k)}}{12\nu AJ_\perp I^{(k)}}\frac{\partial f_0}{\partial\psi}. \tag{20}$$

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Since at the boundaries between different classes of trapped particles, Eq. (3), the quantities $H^{(k)}$ satisfy

$$H_+^{(k)}=H_-^{(k)}+H_-^{(k+1)}, \tag{21}$$

and $H^{(k)}=0$ at the bottoms of the magnetic wells (10), the boundary conditions (9) and (11) are satisfied by (20).

Now let us consider the expression for the particle flux density of trapped particles across the magnetic surface $\psi=$const with area S,

$$F_n=\frac{\pi}{mS}\oint_\psi \frac{\mathrm{d}S}{|\nabla\psi|}B\sum_{\sigma=\pm1}\int \mathrm{d}w\int \mathrm{d}J_\perp f_1\frac{\mathbf{v}\cdot\nabla\psi}{|v_\parallel|}. \tag{22}$$

Here, the superscript (k) has been omitted on f_1. Using the fact that $f_1=0$ for passing particles and with the help of (16) we can integrate by parts (22) over $J_\perp$ once,

$$F_n=\frac{2\pi}{3mS}\oint_\psi \frac{\mathrm{d}S}{|\nabla\psi|}\int_{e\phi}^{\infty}\mathrm{d}w\int_{J_{\perp\min}^{(\mathrm{abs})}}^{J_{\perp\max}}\mathrm{d}J_\perp\ |v_\parallel|(3v^2+v_\parallel^2)\times\frac{|\nabla\psi|k_G}{\omega_c}\frac{\partial f_1}{\partial J_\perp}. \tag{23}$$

Here, $J_{\perp\min}^{(\mathrm{abs})}=v^2/B_{\max}^{(\mathrm{abs})}$ corresponds to the trapped–passing boundary and $J_{\perp\max}(\mathbf{x})=v^2/B(\mathbf{x})$. Using (12) we can express the averages over the magnetic surface area, $\langle P\rangle_S=\langle|\nabla\psi|P\rangle/\langle|\nabla\psi|\rangle$, through the integrals along the magnetic field line. Thus we have for the flux density,

$$\begin{aligned}F_n&=\frac{2\pi}{3m}\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}|\nabla\psi|\right)^{-1}\\&\quad\times\int_0^{L_s}\frac{\mathrm{d}s}{B}\int_{e\phi}^{\infty}\mathrm{d}w\int_{J_{\perp\min}^{(\mathrm{abs})}}^{J_{\perp\max}}\mathrm{d}J_\perp|v_\parallel|(3v^2+v_\parallel^2)\frac{|\nabla\psi|k_G}{\omega_c}\frac{\partial f_1}{\partial J_\perp}\\&=\frac{2\pi}{3m}\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}|\nabla\psi|\right)^{-1}\int_{e\phi}^{\infty}\mathrm{d}w\int_{J_{\perp\min}^{(\mathrm{abs})}}^{J_{\perp\max}^{(\mathrm{abs})}}\mathrm{d}J_\perp\\&\quad\times\sum_{j=1}^{j_{\max}}\int_{s_j^{(\min)}}^{s_j^{(\max)}}\mathrm{d}s|v_\parallel|(3v^2+v_\parallel^2)\frac{|\nabla\psi|k_G}{B\omega_c}\frac{\partial f_1}{\partial J_\perp}.\end{aligned} \tag{24}$$

Here, $J_{\perp\max}^{(\mathrm{abs})}=v^2/B_{\min}^{(\mathrm{abs})}$ where $B_{\min}^{(\mathrm{abs})}$ is the minimum magnetic field module value on a given magnetic surface. In the last expression of (24), the order of integration along the field line and over the velocity space has been changed. The index j numbers now the intervals $[s_j^{(\min)},s_j^{(\max)}]$where $v_\parallel^2>0$ for given $J_\perp$ and w. For different j these intervals include, in general, a different number of magnetic field minima. Note that the ends of the intervals not necessarily are reflection points because for some small fraction of trapped particles the reflection points are outside the interval $[0,L_s]$. The fraction of such particles, however, becomes vanishingly small with $L_s\to\infty$. Since the index j is different from the ripple index $k=k(j,J_\perp,w)$ the notation for the integration limits has been changed, $s_j^{(\min)}=s_{\min,k}$, $s_j^{(\max)}=s_{\max,k}$. Substituting now in (24) the derivative of the distribution function from (20) and using (19) we obtain

$$F_n=-\frac{\pi}{36m\nu}\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}|\nabla\psi|\right)^{-1}\int_{e\phi}^{\infty}\mathrm{d}w\int_{J_{\perp\min}^{(\mathrm{abs})}}^{J_{\perp\max}^{(\mathrm{abs})}}\mathrm{d}J_\perp\times\sum_{j=1}^{j_{\max}}\frac{[H^{(k)}]^2}{AJ_\perp I^{(k)}}\frac{\partial f_0}{\partial\psi}. \tag{25}$$

Introducing instead of $J_\perp$ the new integration variable $b'=v^2/(J_\perp B_0)$ where B_0 is some reference magnetic field, we finally obtain

$$F_n=-\frac{\sqrt{8}}{9\pi^{3/2}}\frac{v_T^2\rho_L^2}{\nu R^2}\epsilon_{\mathrm{eff}}^{3/2}\int_0^{\infty}\frac{\mathrm{d}z\,e^{-z}z^{5/2}}{A(z)}\frac{n}{f_0}\frac{\partial f_0}{\partial r}. \tag{26}$$

Here $v_T=\sqrt{2T/m}$ is the thermal velocity, $\rho_{\mathrm{L}}=mcv_T/(eB_0)$ is the mean Larmor radius, R is the big radius of the geometrical axis, $w=Tz+e\phi$ and $n=n(\psi)$ is a particle density. We used the following notation for the averaged normal derivative:

$$\frac{\partial f_0}{\partial r}\equiv\left\langle\frac{\partial f_0}{\partial n_\psi}\right\rangle=\frac{\partial f_0}{\partial\psi}\langle|\nabla\psi|\rangle, \tag{27}$$

where average $\langle\cdots\rangle$ is defined by (12). One also can view such a normal derivative as a derivative over the effective radius $r=r(\psi)$ defined through the differential relation between the volume $V(\psi)$ enclosed by the magnetic surface and the magnetic surface area $S(\psi)$,

$$\mathrm{d}r\equiv\frac{\mathrm{d}V}{S}=\frac{\mathrm{d}\psi}{\langle|\nabla\psi|\rangle}. \tag{28}$$

The equation for the energy flux density is obtained in a similar way and differs from (26) by a factor zT in the subintegrand.

The only difference of (26) from the corresponding formula for the standard stellarator [see Eq. (2.16) of Ref. 2] is the replacement of the helical modulation amplitude ϵ_h by the effective ripple modulation amplitude ϵ_{eff} given by

$$\epsilon_{\mathrm{eff}}^{3/2}=\frac{\pi R^2}{8\sqrt{2}}\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{ds}{B}\right)\times\left(\int_0^{L_s}\frac{ds}{B}|\nabla\psi|\right)^{-2}\int_{B_{\min}^{(\mathrm{abs})}/B_0}^{B_{\max}^{(\mathrm{abs})}/B_0}\mathrm{d}b'\sum_{j=1}^{j_{\max}}\frac{\hat{H}_j^2}{\hat{I}_j}, \tag{29}$$

with

$$\hat{H}_j=\frac{1}{b'}\int_{s_j^{(\min)}}^{s_j^{(\max)}}\frac{\mathrm{d}s}{B}\sqrt{b'-\frac{B}{B_0}}\left(4\frac{B_0}{B}-\frac{1}{b'}\right)|\nabla\psi|k_G, \tag{30}$$

$$\hat{I}_j=\int_{s_j^{(\min)}}^{s_j^{(\max)}}\frac{\mathrm{d}s}{B}\sqrt{1-\frac{B}{B_0b'}}. \tag{31}$$

Often instead of the Lorentz collision operator (2) an approximate collision operator with diffusion over parallel velocity is used,

$$\mathrm{St}_\parallel(f)=\nu A(v_\perp^2)v_\perp^2\frac{\partial^2 f}{\partial v_\parallel^2}. \tag{32}$$

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The effective ripple amplitude $\epsilon_{\rm eff}$ in this case is again given by (29) where $\hat{H}_j$ and $\hat{I}_j$ should be replaced with $\hat{G}_j$ and $\hat{J}_j$, respectively,

$$\hat{G}_j=\int_{s_j^{(\min)}}^{s_j^{(\max)}}\frac{\mathrm{d}s}{B_0}\sqrt{b'-\frac{B}{B_0}}\left(1+2b'\frac{B_0}{B}\right)|\nabla\psi|k_G, \tag{33}$$

$$\hat{J}_j=\int_{s_j^{(\min)}}^{s_j^{(\max)}}\frac{\mathrm{d}s}{B_0}\sqrt{b'-\frac{B}{B_0}}. \tag{34}$$

It can be checked that for the field parameters of the standard stellarator the formulas for $\epsilon_{\rm eff}^{3/2}$ approximately yield $\epsilon_h^{3/2}$.

Formula (29) includes the contributions from all classes of trapped particles. These contributions can be formally separated,

$$\epsilon_{\rm eff}^{3/2}=\sum_{i=1}^{\infty}\epsilon_{{\rm eff},i}^{3/2}, \tag{35}$$

where index i stands for the class of particles trapped within i wells and $\epsilon_{{\rm eff},i}^{3/2}$ is given by (29) with the additional factor $\delta_{i,\mathcal{I}}$ in the sum over j. Here $\delta_{i,k}$ is the Kronecker symbol and $\mathcal{I}=\mathcal{I}(j,b')$ is the number of the magnetic field minima in the interval $[s_j^{(\min)},s_j^{(\max)}]$. The quantities $\epsilon_{{\rm eff},i}^{3/2}$ can be considered only as a qualitative estimate of the role of different particle classes in $1/\nu$ transport since the classes are strongly linked by Coulomb diffusion in phase space.

Considering the unperturbed particle density averaged over the volume between the neighboring magnetic surfaces and following the lines of derivation of (24) we can calculate the fractions of particles belonging to different classes,

$$\begin{aligned}\langle n\rangle&=\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}\right)^{-1}\int_0^{L_s}\frac{\mathrm{d}s}{B}\sum_{\sigma=\pm1}\int\mathrm{d}w\int\mathrm{d}J_\perp\frac{\pi Bf_0}{m|v_\parallel|}\\&=n\left(\sum_{i=1}^{\infty}c_i^{(\rm tr)}+c^{(p)}\right),\end{aligned} \tag{36}$$

where

$$\begin{aligned}c_i^{(tr)}&=\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}\right)^{-1}\int_{B_{\min}^{(\rm abs)}/B_0}^{B_{\max}^{(\rm abs)}/B_0}\mathrm{d}b'\\&\quad\times\sum_{j=1}^{j_{\max}}\frac{\delta_{i,\mathcal{I}}}{2B_0b'^2}\int_{s_j^{(\min)}}^{s_j^{(\max)}}\mathrm{d}s\left(1-\frac{B}{B_0b'}\right)^{-1/2}\\&=\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}\right)^{-1}\sum_{j=1}^{j_{\max}}(\hat{I}_j(b'_{\max}(i,j))-\hat{I}_j(b'_{\min}(i,j)))\end{aligned} \tag{37}$$

is the fraction of particles trapped in i minima. Here $\hat{I}_j$ is given by (31) and $b'_{\max}(i,j)$ and $b'_{\min}(i,j)$ are maximum and minimum values of the variable b' which satisfy $\mathcal{I}(j,b')=i$, respectively [clearly $\hat{I}_j(b'_{\min}(1,j))=0$]. The fraction of passing particles $c^{(p)}$ is given by

$$c^{(p)}=1-\lim_{L_s\to\infty}\left(\int_0^{L_s}\frac{\mathrm{d}s}{B}\right)^{-1}\int_0^{L_s}\frac{\mathrm{d}s}{B}\sqrt{1-\frac{B}{B_{\max}^{(\rm abs)}}}. \tag{38}$$

The total sum in the parentheses of (36) equals 1 (for a Maxwellian distribution the density is constant on the magnetic surface). The equivalent amplitudes of ripple and toroidal modulation defined as

$$\epsilon_{he}=\frac{\pi^2}{8}(c_1^{(\rm tr)})^2,\quad \epsilon_{te}=\frac{\pi^2}{8}(1-c^{(p)})^2, \tag{39}$$

are useful for the comparison of the results. For the standard stellarator $\epsilon_{he}=\epsilon_h$.

B. Check for quasi-helically symmetric systems and results in Boozer coordinates

The quasi-helically symmetric systems[14,17,18] are defined as those systems in which the magnetic field module expressed as a function of Boozer coordinates ψ,θ,ζ depends only on the combination of the poloidal and toroidal angle-like variables, $B=B(\psi,m\theta-n\zeta)$, where m and n are integer numbers. Using the coordinates $\mathbf{x}$ introduced in the previous section where $\theta_0=\theta-\iota\zeta$ and ι is the rotational transform angle, we obtain

$$\begin{aligned}|\nabla\psi|k_G&=\frac{1}{B}(\mathbf{h}\times\nabla B)\cdot\nabla\psi\\&=\frac{1}{B\sqrt{g}}\left(h_2\frac{\partial B}{\partial x^3}-h_3\frac{\partial B}{\partial x^2}\right)\\&=\frac{I}{J+\iota I}\frac{\partial B}{\partial\zeta}-\frac{\partial B}{\partial\theta_0}.\end{aligned} \tag{40}$$

Here $h_2=I/B$ and $h_3=(J+\iota I)/B$ are the covariant components of $\mathbf{h}$ where I and J denote the covariant components of the magnetic field in Boozer coordinates $B_\theta=I(\psi)$, $B_\zeta=J(\psi)$ which are constant on the magnetic surface. Here it was used that the Jacobian of coordinates $\mathbf{x}$ is the same as the Jacobian of Boozer coordinates $\sqrt{g}=(J+\iota I)/B^2$. Using the fact that $\sqrt{g_{33}}=(J+\iota I)/B$ for coordinates $\mathbf{x}$, we obtain from (30) for $\hat{H}_j$,

$$\begin{aligned}\hat{H}_j&=\frac{1}{b'}\int_{\zeta_j^{(\min)}}^{\zeta_j^{(\max)}}\frac{\mathrm{d}\zeta}{B^2}\sqrt{b'-\frac{B}{B_0}}\left(4\frac{B_0}{B}-\frac{1}{b'}\right)\\&\quad\times\left(I\frac{\partial B}{\partial\zeta}-(J+\iota I)\frac{\partial B}{\partial\theta_0}\right).\end{aligned} \tag{41}$$

For quasi-helical symmetry the sub-integrand expression becomes a complete derivative over ζ,

$$\hat{H}_j=\frac{nI+mJ}{n-\iota m}\frac{1}{b'}\int_{\zeta_j^{(\min)}}^{\zeta_j^{(\max)}}\frac{\mathrm{d}\zeta}{B^2}\sqrt{b'-\frac{B}{B_0}}\left(4\frac{B_0}{B}-\frac{1}{b'}\right)\frac{\partial B}{\partial\zeta}=0. \tag{42}$$

Using (33) one can show in a similar way that $\hat{G}_j=0$ for quasi-helically symmetric systems.

Finally, it is also useful to present (29) in Boozer coordinates,

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$$\epsilon_{\rm eff}^{3/2}=\frac{\pi R^2}{8\sqrt{2}}\lim_{L_\zeta\to\infty}\left(\int_0^{L_\zeta}\frac{d\zeta}{B^2}\right)\left(\int_0^{L_\zeta}\frac{d\zeta}{B^2}\sqrt{g^{11}}\right)^{-2}\times\int_{B_{\min}^{(\rm abs)}/B_0}^{B_{\max}^{(\rm abs)}/B_0}{\rm d}b'\sum_{j=1}^{j_{\max}}\frac{H_{fj}^2}{I_{fj}},\tag{43}$$

where

$$H_{fj}=\frac{\hat{H}_j}{J+\iota I}=\frac{1}{(J+\iota I)b'}\times\int_{\zeta_j^{(\min)}}^{\zeta_j^{(\max)}}\frac{{\rm d}\zeta}{B^2}\sqrt{b'-\frac{B}{B_0}}\left(4\frac{B_0}{B}-\frac{1}{b'}\right)\left(I\frac{\partial B}{\partial\zeta}-J\frac{\partial B}{\partial\theta}\right),\tag{44}$$

$$I_{fj}=\frac{\hat{I}_j}{J+\iota I}=\int_{\zeta_j^{(\min)}}^{\zeta_j^{(\max)}}\frac{{\rm d}\zeta}{B^2}\sqrt{1-\frac{B}{B_0b'}},\tag{45}$$

and $g^{11}=(\nabla\psi)^2$ is the metric tensor component. The integration in Eqs. (43)–(45) is along the field lines, $\theta=\theta_0+\iota\zeta$. Such a representation may be useful, in particular, for the comparison of the results with analytical works employing simplified magnetic field models (see, e.g., Ref. 19).

III. MAGNETIC FIELD REPRESENTATION

The technique derived in the previous section is used to compute the $1/\nu$ diffusion in three different configurations: (i) a simplified U-3M configuration; (ii) the original Helias configuration;[13] and quasi-helically symmetric configuration.[14] It is well known that Helias configurations have been further optimized with respect to various properties[20] and that the transport properties of the late versions have been studied extensively.[21,8]

However, to demonstrate the capability of the method developed in this paper, the three examples listed above have been chosen for the following reasons: (i) the neoclassical properties of U-3M are very close to those of the standard stellarator with $l=3$; (ii) the original Helias configuration has been optimized only with respect to MHD properties and has significant asymmetric losses. On the other hand, the neoclassical properties of this configuration are rather different from a standard stellarator; (iii) the QHS stellarator is an extreme case with very small asymmetric losses. The fairly big asymmetric losses of the first two configurations made it easier to calculate the $1/\nu$-diffusion coefficients with the help of Monte Carlo calculation and to benchmark the method. For the third configuration this would be very difficult because the symmetric losses are dominant and because of the fact that the calculations yield for the overall diffusion.

For these examples we consider the zero β case and, in contrast to Refs. 13 and 14, the Helias and QHS stellarator magnetic fields are represented in real space as a superposition of toroidal harmonic functions containing the associated Legendre functions in the form[11]

$$\mathbf{B}=\nabla\Phi+\nabla\times\mathbf{A}_\varphi+\mathbf{e}_\varphi b_0R/\rho,\tag{46}$$

$$\Phi=\sum_{n=0}^{\infty}\sum_{\substack{m=-\infty\\(m\neq0)}}^{\infty}H_{nm}\Phi_{nm},\tag{47}$$

$$\Phi_{nm}=\sqrt{\cosh\eta-\cos\theta}\,Q_{n-1/2}^m(\cosh\eta)\sin(n\theta+m\varphi),\tag{48}$$

$$A_\varphi=\sum_{n=0}^{\infty}C_nA_{\varphi n},\tag{49}$$

$$A_{\varphi n}=\sqrt{\cosh\eta-\cos\theta}\,Q_{n-1/2}^1(\cosh\eta)\cos(n\theta).\tag{50}$$

Here $Q_{n-1/2}^m$ are associated Legendre functions of the second kind, (η,θ,φ) is the toroidal system of coordinates associated with the cylindrical system (ρ,φ,z), n and m are the poloidal and toroidal harmonic numbers, R is the major radius of the torus, and H_{nm} and C_n are the expansion coefficients.

Sets of decomposition terms are used that satisfy the conditions

$$|n|\leqslant12,\quad m=m_pM,\quad 0\leqslant M\leqslant12,\tag{51}$$

where M is the number of toroidal harmonic within one magnetic field period, m_p is the number of periods along the torus ($m_p=5$ for the original Helias configuration,[13] $m_p=6$ for the QHS stellarator[14]).

The expansion coefficients of the superpositions are obtained by minimizing the magnetic field component that is normal to the prescribed boundaries with equations presented in Refs. 13 and 14.

For the U-3M $l=3$ torsatron calculations we consider a simplified magnetic field which contains only one toroidal harmonic in (47) (with $n=3$, $m=9$, $R=100$ cm).

For a representation of the original Helias magnetic configuration ($m_p=5$), the magnetic surfaces are obtained by a straightforward numerical field line integration. Two magnetic surfaces together with the boundary surface[13] in the cross-sections $\varphi=0$ and $\varphi=\pi/5$ (one half of the magnetic field period) are shown in Fig. 2 (in cylindrical ρ,φ,z coordinates, the sizes are given in the same dimensionless form normalized with respect to the mean plasma radius defined in Ref. 13). The magnetic surfaces are characterized by the initial point of the integration ($\rho=\rho_0$, $\varphi=0$, $z=0$). The surface $\rho_0=13.16$ is near the boundary surface, the surface $\rho_0=12.9$ is near the half distance from the magnetic axis. Numbers indicate the magnetic field lines for which further calculations are performed.

Figure 3 shows the distribution of the magnetic field module along some field lines on the surface $\rho_0=13.16$. A rather large magnetic field modulation along the field lines can be seen.

The magnetic configuration of the QHS stellarator for the above magnetic field representation has been discussed in Refs. 11 and 16.

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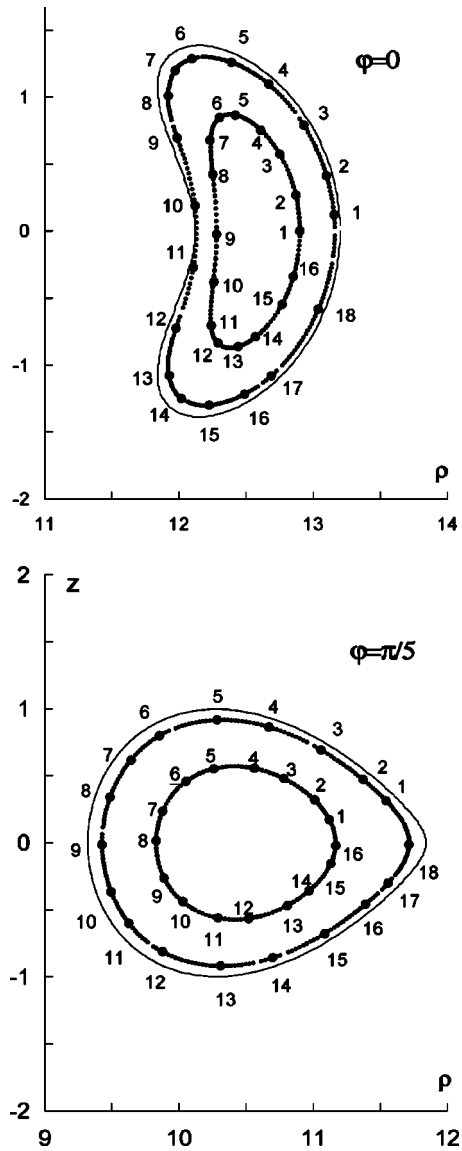

FIG. 2. Boundary surface (Ref. 13) (solid line) and two magnetic surfaces: near boundary ($\rho_0=13.16$, $\iota=0.717$) and inner ($\rho_0=12.9$, $\iota=0.59$), ι is a rotational transform (in units of 2π); numbers show magnetic field lines for which the results of calculations are presented in Figs. 3 and 4.

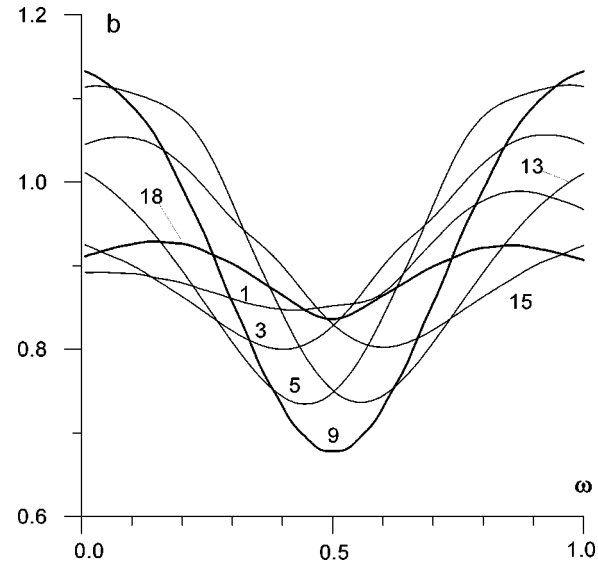

FIG. 3. The distribution of B within one magnetic field period for some of the field lines numbered in Fig. 2 for the surface $\rho_0=13.16$ ($b=B/B_0, \omega=m_p\varphi/2\pi$).

IV. BOUNCE-AVERAGED TRAPPED PARTICLE DRIFT FOR THE ORIGINAL HELIAS CONFIGURATION

Before calculating the neoclassical transport let us consider trapped particle drift in the original Helias configuration[13] in order to demonstrate the significant difference of particle drift in this configuration compared to the standard stellarator and to clarify some features of the neoclassical transport coefficients obtained in the following sections. The calculated results for the bounce-averaged trapped particle drift velocity are compared to analogous results obtained for the QHS stellarator in Refs. 11 and 16.

A. Bounce-averaged trapped particle drift velocity

In this section the bounce-averaged trapped particle drift velocity is analyzed with the help of the technique developed in Refs. 11 and 16. Briefly, the technique consists in a calculation of the bounce period displacement $\delta\psi$ given by (6).

The results of these calculations are represented by the normalized quantities η and γ defined as

$$\eta=v_{an}\frac{eR}{mcJ_\perp}, \quad v_{an}=\frac{\delta\psi}{\tau_b|\nabla\psi_i|}, \tag{52}$$

$$\gamma=v_{\|i}/v_{\perp 0}, \tag{53}$$

where v_{an} is the bounce-averaged drift velocity of the trapped particles across the magnetic surface, $2eR/mcJ_\perp$ is the inverse amplitude of the bounce-averaged drift velocity for the associated standard stellarator, $\nabla\psi_i$ and $v_{\|i}$ are the corresponding values at the initial point of integration (minimum B point), τ_b is the bounce period, $v_{\perp 0}=\sqrt{J_\perp B_0}$, B_0 is the reference toroidal magnetic field for the configuration represented by the mean magnetic field on the circular axis of the torus. Minimum B points and the $\nabla\psi_i$ values are obtained together with the computation of the magnetic configuration.

The γ parameter is directly related to the depth of particle trapping. For deeply trapped particles $v_{\|i}=0$ and $\gamma=0$ whereas at the transition from trapped particles to passing $v_{\|i}=v_{\|i\max}$ and $\gamma=\gamma_{\max}$.

For the standard stellarator $\eta=\eta_m\sin\vartheta$ where ϑ is the poloidal angle of quasi-toroidal coordinates and $\eta_m=0.5$. The relations (52) for η and v_{an} allow a convenient comparison of the quantities v_{an} in various magnetic configurations of equivalent sizes.

B. Results for the bounce-averaged drift velocity

The results of the η calculation are presented below for the magnetic surfaces shown in Fig. 2. In order to illustrate the choice of the initial points for the $\delta\psi$ integration [see Eq. (52)] the B/B_0 distributions within a single magnetic field period are shown in Fig. 3 for several field lines on the $\rho_0=13.16$ magnetic surface which is located near the boundary. The results for the normalized drift velocities η as func-

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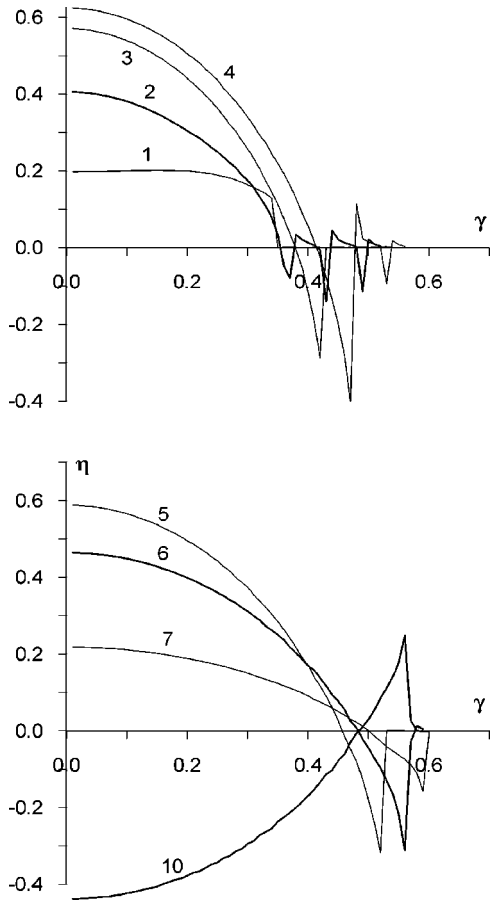

FIG. 4. Parameter η, Eq. (52), as a function of γ, Eq. (53), for the magnetic surface $\rho_0=12.9$ (the curves are numbered according to numbers of field lines in Fig. 2; for field lines 9, 10, 11, 12, 13, 14 and 15 the η values are approximately equal in magnitude but opposite in sign to those in the graphs for field lines 7, 6, 5, 4, 3, 2 and 1, respectively; for field lines 8 and 16 $\eta\approx 0$ independent of γ; $\gamma_t^{(1)}=0.34$, $\gamma_t^{(2)}=0.36$, $\gamma_t^{(3)}=0.42$, $\gamma_t^{(4)}=0.47$, $\gamma_t^{(5)}=0.52$, $\gamma_t^{(6)}=0.56$, $\gamma_t^{(7)}=0.59$.

tions of γ are presented in Fig. 4 for the $\rho_0=12.9$ magnetic surface. The analogous results for the $\rho_0=13.16$ magnetic surface are not essentially different from those in Fig. 4.

It can be observed that in the interval $0\leqslant\gamma\leqslant\gamma_{\max}$ which corresponds to the trapped particles ($\gamma_{\max}=0.6-0.67$ for $\rho_0=13.16$ and $\gamma_{\max}=0.55-0.6$ for $\rho_0=12.9$) the η values vary with γ in contrast to the standard stellarator model for which η is independent of γ.

In the range $0\leqslant\gamma\leqslant\gamma_{\max}$ there exist particular γ values corresponding to transitions between different classes of particles (trapped within a single magnetic field period, passing or trapped within a few magnetic field periods). These γ values are given in the figure captions and denoted by $\gamma_t^{(n)}$ (n is the field line index in Fig. 2).

For deeply trapped particles ($\gamma\approx 0$) the maximum η value $\eta_m\approx 0.62$ (for $\rho_0=13.16$ and for $\rho_0=12.9$) and is slightly greater than that for the standard stellarator ($\eta_m=0.5$). At the same time, with γ increasing η is decreasing and passes through zero at a certain value of the trapping depth which is close to the transition boundary from trapped to passing particles.

Such a dependence of η on the trapping depth shows that the bounce-averaged drift velocity for trapped particles across magnetic surfaces in the original Helias is, on average, less than that in the standard stellarator with identical major radius and magnetic field helical inhomogeneity.

Calculations of such a kind were made in Refs. 11 and 16 for the QHS stellarator.[14] It follows from the calculations that the η value also varies with γ but for the QHS stellarator (for the magnetic surface which is close to the boundary) the η value, on average, is approximately ten times less than that for the original Helias. This feature must lead to a significant decrease of the neoclassical transport coefficients due to the magnetic field asymmetry in the QHS stellarator when compared to the original Helias.

V. NEOCLASSICAL TRANSPORT EVALUATION

A. Use of the field line integration technique

In order to perform the neoclassical transport calculation with the help of formula (29) in Sec. II A this formula must be supplemented by the magnetic field line equations,

$$\frac{d\boldsymbol{\xi}^{\mathbf{i}}}{ds}=\frac{B^i}{B}, \tag{54}$$

and the equations for the vector $\mathbf{P}\equiv\nabla\psi$ normal to the magnetic surface (see Ref. 22),

$$\frac{dP_i}{ds}=-\frac{1}{B}\frac{\partial B^j}{\partial\xi^i}P_j\,. \tag{55}$$

Here B^i are the contravariant components of the vector $\mathbf{B}$ in a nonorthogonal curvilinear system of coordinates ξ^i, $P_i=\partial\psi/\partial\xi^i$ are the covariant components of the vector $\mathbf{P}$, and the factor $1/B$ corresponds to the integration variable s. In particular, the cylindrical coordinates ρ,φ,z are taken as the coordinates ξ^i. Equation (55) allows one to find the $\nabla\psi$ value on a given magnetic field line and calculate the field line geodesic curvature which enters Eqs. (30) and (33). Such a method of calculation of the geodesic curvature has been introduced for the calculation of the transport coefficients in the Pfirsch–Schlüter regime in arbitrary stellarator magnetic fields.[12]

The integral over b' in (29) is approximated by the finite difference formula. The values of the sub-integrand in the nodes b_i' ($i=1,2,\ldots$) are calculated using (30) and (31) together with the field line integration (54), integration of the normal (55) and calculation of the other integrals entering (29) within the same Runge–Kutta procedure. For the magnetic configurations discussed in Sec. III (original Helias, QHS stellarator, U-3M) the integration was over 250 magnetic field periods (for nonisland magnetic surfaces). Since most of the computer time is spent for the calculation of the magnetic field, the use of a few hundred grid nodes does not increase the computation time significantly because the same magnetic data are used for all the nodes b_i'.

The calculation results for $\epsilon_{\text{eff}}^{3/2}$ are presented in Table I and in Fig. 5 in form of $\epsilon_{\text{L}}^{3/2}$ and $\epsilon_{\parallel}^{3/2}$. The quantity $\epsilon_{\text{L}}^{3/2}$ is calculated with help of Eqs. (30) and (31) and corresponds to the Lorentz collision operator. The quantity $\epsilon_{\parallel}^{3/2}$ corresponds to the approximate collision operator (32) (small $v_{\parallel}/v$ ratio

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TABLE I. Results of $\epsilon_{\rm eff}^{3/2}$ calculations.

Config.	$\overline{\epsilon_h}$	$\overline{\epsilon_h^{3/2}}$	$\epsilon_L^{3/2}$	Notes
Helias	0.124	4.36×10^{-2}	2.9×10^{-2}	$\rho_0=12.59$
Helias	0.124	4.36×10^{-2}	2.9×10^{-2}	$\rho_0=12.9$
Helias	0.139	5.19×10^{-2}	3.0×10^{-2}	$\rho_0=13.16$
QHS	0.015	1.78×10^{-3}	1.1×10^{-5}	$\rho_0=11.95$
QHS	0.063	1.54×10^{-2}	7.0×10^{-5}	$\rho_0=12.14$
QHS	0.10	3.15×10^{-2}	1.6×10^{-4}	$\rho_0=12.25$
QHS	0.12	4.15×10^{-2}	4.1×10^{-4}	$\rho_0=12.33$
U-3M	0.048	1.07×10^{-2}	1.4×10^{-2}	$\iota\approx0.405$

approximation) with Eqs. (33) and (34) used for $\epsilon_{\rm eff}^{3/2}$. Besides the quantities $\epsilon_\parallel^{3/2}$ and $\epsilon_{\rm L}^{3/2}$ an average ϵ_h quantity, $\overline{\epsilon_h}$ is presented in Table 1 [$\overline{\epsilon_h}=(\epsilon_{h\max}+\epsilon_{h\min})/2$, $\epsilon_{h\max}$ and $\epsilon_{h\min}$ are the maximum and minimum values of ripple modulation amplitude, respectively]. The data in the last column indicate the magnetic surface for which the calculation is made.

The calculations performed have shown that for the sufficiently small ϵ_h (in particular for U-3M) the results for $\epsilon_{\rm eff}^{3/2}$ obtained with the use of the Lorentz collision operator ($\epsilon_{\rm eff}^{3/2}=\epsilon_{\rm L}^{3/2}$) do not essentially differ from those obtained with the use of the collision operator in the small $\nu_\parallel/\nu$ ratio approximation ($\epsilon_{\rm eff}^{3/2}=\epsilon_\parallel^{3/2}$). However, with increasing ϵ_h a marked difference in the results appears and the approximation turns out to be inappropriate. For example, for the original Helias $\epsilon_\parallel^{3/2}=1.2\times10^{-2}$ (for $\rho_0=12.9$) and for QHS stellarator $\epsilon_\parallel^{3/2}=2.0\times10^{-4}$ (for $\rho_0=12.33$). These values are approximately two times smaller than the corresponding values for $\epsilon_{\rm L}^{3/2}$ (see Table I).

The results show that for U-3M $\epsilon_{\rm eff}^{3/2}$ is slightly greater than $\overline{\epsilon_h^{3/2}}$ whereas for the original Helias and QHS configuration $\epsilon_{\rm eff}^{3/2}$ is less than $\overline{\epsilon_h^{3/2}}$. For the original Helias such a decrease in $\epsilon_{\rm eff}^{3/2}$ (demonstrating the advantage of the original Helias to the standard stellarator with identical R and $\overline{\epsilon_h}$) can be explained by decreasing the quantity v_{an} (or η, see Eq. (52)) when the parameter γ tends to the transition from trapped (within one magnetic field period) particles to those which are passing or trapped within several magnetic field periods (see Sec. IV). Consequently, the estimate of the multiply trapped particle contribution to the $1/\nu$ transport, $\epsilon_{\rm eff}^{3/2}-\epsilon_{\rm eff,1}^{3/2}$ [see (35)] shows that this contribution is less than 1%.

For the QHS configuration considered here the decrease in $\epsilon_{\rm eff}^{3/2}$ (with respect to $\overline{\epsilon_h^{3/2}}$) turns out to be much stronger, by two orders of magnitude. This behavior is directly related to the quasi-symmetry properties of the magnetic field and manifests itself in a significant decrease of the bounce-averaged trapped particle drift velocity across the magnetic surfaces (the quantities v_{an} and η) and the neoclassical transport coefficients.

Note that for the magnetic field representation of the QHS configuration considered here, quasi-helical symmetry is not realized exactly ($\epsilon_{\rm eff}^{3/2}$ differs from zero; see Sec. II B). Nevertheless, since it appears that $\epsilon_{\rm L}^{3/2}\ll\overline{\epsilon_h^{3/2}}$ the quasi-helical symmetry is only slightly violated. The measure of such a violation is given by the ratio $\epsilon_{\rm L}^{3/2}/\overline{\epsilon_h^{3/2}}$. This ratio varies from 0.005 for the inner magnetic surfaces to 0.01 for the near boundary surface. Nevertheless, $\epsilon_{\rm eff}^{3/2}$ strongly decreases towards the center since $\overline{\epsilon_h}$ is decreasing with the effective radius.

Thus, it follows from the results that for identical sizes of the magnetic systems the neoclassical transport coefficients for the QHS stellarator are significantly smaller (two orders in magnitude) than the corresponding coefficients for the original Helias and U-3M.

From Fig. 5 one can see that for the inner U-3M magnetic surfaces $\epsilon_\parallel^{3/2}$ is essentially less than that for the near boundary region. This fact is explained by a significant decrease in ϵ_h with decreasing distance from the magnetic axis in U-3M. This can be seen in Fig. 6 where the equivalent ϵ_h values, ϵ_{he}, defined by (39) are shown. At the same time, the $\epsilon_{\rm eff}^{3/2}/\overline{\epsilon_h^{3/2}}$ (or $\epsilon_{\rm eff}^{3/2}/\epsilon_{he}^{3/2}$) ratio turns out to be greater for U-3M than that for the original Helias where changes in ϵ_h are rather small.

In contrast to Table I, where $\overline{\epsilon_h}$ is determined through the magnetic field helical modulation along a magnetic field line, ϵ_h, Fig. 6 shows equivalent amplitudes of ripple and toroidal modulations, ϵ_{he} and ϵ_{te}, as obtained from the trapped particle fractions with the use of (39). It can be seen that the approximate quantities $\overline{\epsilon_h}$ in Table I are fairly close to the corresponding quantities ϵ_{he} in Fig. 6.

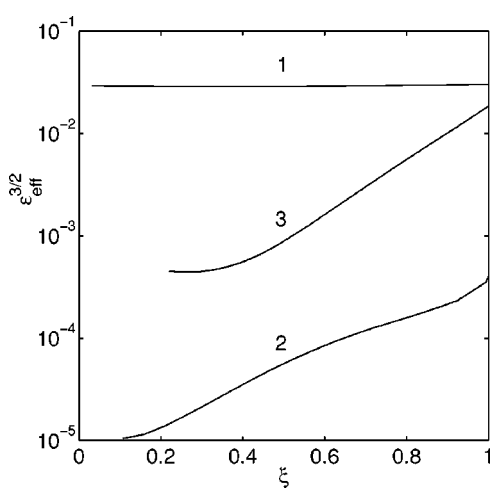

FIG. 5. Parameter $\epsilon_{\rm L}^{3/2}$ for the original Helias (curve 1) and QHS stellarator (curve 2) and parameter $\epsilon_\parallel^{3/2}$ for U-3M (curve 3); ξ is a dimensionless distance from the magnetic axis ($\xi=1$ corresponds to the boundaries).

B. Monte-Carlo calculations

To compare the results, the diffusion coefficients for the $1/\nu$ transport regime are simulated with a conventional Monte-Carlo (MC) method[6] for the original Helias and U-3M configurations which have asymmetric losses bigger than the QHS configuration. For this purpose, the conven-

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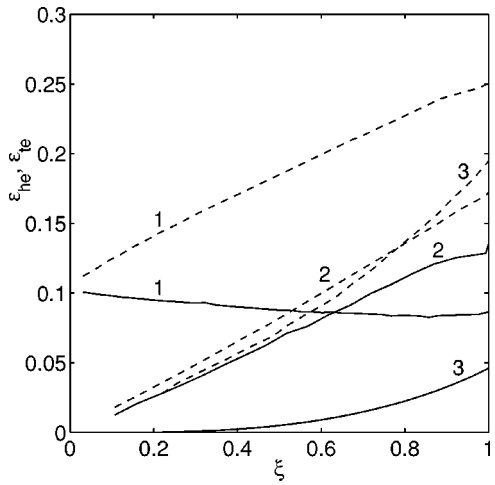

FIG. 6. Equivalent amplitudes of ripple and ''toroidal'' modulation as given by Eqs. (39) for various stellarator configurations: the original Helias (curves 1), QHS stellarator (curves 2), U-3M (curves 3); solid lines for ϵ_{he}, dashed lines for ϵ_{te}, ξ is a dimensionless distance from the magnetic axis.

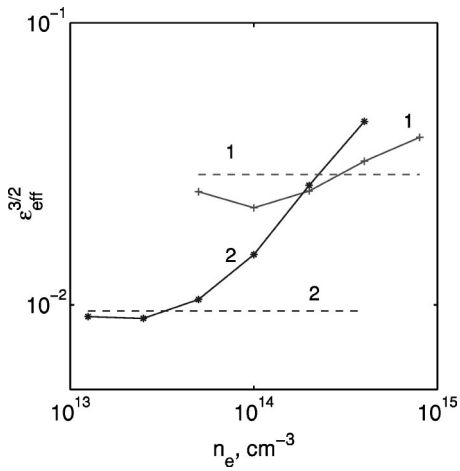

FIG. 7. Normalized results of MC calculations (solid lines) for the original Helias ($\rho_0=12.9$, curves 1) and U-3M ($\rho_0=108$ cm, curves 2); dashed lines show the level of $\epsilon_{\rm eff}^{3/2}$ calculated with the integration technique along the field line.

tional real space coordinate MC code is used, which has been developed for benchmarking the MC solver based on the stochastic mapping technique.[23] Ensembles of 100 and 1000 test particles are taken for the original Helias and U-3M, respectively. The integration of the stochastic trajectories for each particle is performed during the time τ_0 of the order of three Coulomb collision times and the diffusion coefficients are found from $D_\perp=\langle\Delta r^2\rangle/2\tau_0$ where $\langle\Delta r^2\rangle$ is the dispersion of the effective radius (28). The corresponding values of $\epsilon_{\rm eff}^{3/2}$ are obtained using the expression for the diffusion coefficient $D_\perp$ which easily follows from (26) and the calculated $D_\perp$ values.

For the MC calculations for the original Helias, a configuration is considered which corresponds to $R=1125$ cm and a starting surface $\rho_0=1290$ cm ($R=11.25$, $\rho_0=12.9$ in the dimensionless form). The calculations were performed for electrons with the following plasma parameters: $B=50$ kG, $T=10$ keV, $n=(5\times10^{13}-8\times10^{14})$ cm^{-3}.

For the MC calculations in U-3M we consider the following plasma parameters (for electrons): $B=300$ kG, $T=3$ keV, $n=(1.25\times10^{13}-4\times10^{14})$ cm^{-3}. Due to a small rotational transform in the central region of the U-3M configuration, the $1/\nu$ regime can play the dominant role only at the edge of the confinement region. For this reason, MC calculations are performed only in the region near the boundary and the parameters are chosen to satisfy the condition of small particle deviations from the starting magnetic surface ($\rho_0=108$ cm) during the time τ_0 in order to avoid the escape of test particles through the confinement region boundary.

The results of MC $\epsilon_{\rm eff}^{3/2}$ calculations for the original Helias and U-3M are presented in Fig. 7. From Fig. 7 it follows that for the original Helias the MC calculation results are in a satisfactory agreement with the $\epsilon_{\rm L}^{3/2}$ value presented in Table I. For U-3M the results of MC calculations are in good agreement with the $\epsilon_{\parallel}^{3/2}$ value (see Fig. 5) for $n=(1.25-8)\times10^{13}$ cm^{-3}. A noticeable difference in the results for higher n can be explained by the fact that $\epsilon_{\parallel}^{3/2}$ corresponds to the transport associated with the magnetic field asymmetry only whereas the MC calculations include also axisymmetric and helically symmetric parts of the transport which manifest themselves more strongly with increasing collision frequency.

VI. CONCLUSIONS

In the present paper a technique is proposed which allows us to calculate the local transport coefficients for the $1/\nu$ regime using an integration along magnetic field lines. This technique can be applied to any stellarator magnetic field in regions where regular or island magnetic surfaces exist. Within this method the transport coefficients are presented in a standard form containing a factor $\epsilon_{\rm eff}^{3/2}$ depending on the magnetic field geometry. This factor naturally takes into account the contributions to the $1/\nu$ transport coming from all classes of trapped particles, i.e., particles trapped not only within one magnetic field period but also within several magnetic field periods. Note that the necessity of taking into account particles trapped in more than one ripple well in some magnetic configurations was pointed out in Ref. 24. where an attempt of such an account has been undertaken.

Using the technique given in the present paper, calculations can be performed in real-space coordinates or in magnetic coordinates. If the magnetic field is originally available in real-space coordinates, calculations can be performed without a field transformation to magnetic coordinates. The computer time needed for the computation of transport coefficients practically does not exceed the time for computing the magnetic surface of interest using the integration of the magnetic field line equations. The Monte-Carlo calculations for both the simplified U-3M and the original Helias show satisfactory agreement with the field line integration technique for plasma parameters corresponding to the $1/\nu$ regime. In particular, the method yields zero transport coefficients for symmetric configurations and configurations with exact quasi-helical symmetry.

The proposed technique has been applied to study the $1/\nu$ transport in the Helias-type magnetic configurations

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[original Helias[13] and quasi-helically symmetric (QHS) stellarator[14]] and simplified U-3M configuration in the zero-β limit. In particular, it was shown that for the QHS stellarator[14] the contribution of the magnetic field asymmetry to the neoclassical transport coefficients turns out to be approximately 100 times less than the corresponding contribution in the original Helias[13] with nonoptimized transport properties. This result is in good agreement with the corresponding estimate made in Ref. 11. From the calculations it also follows that for inner magnetic surfaces of the QHS stellarator some violation of quasi-helical symmetry does not aggravate the confinement properties in the $1/\nu$ regime because $\epsilon_{\mathrm{eff}}^{3/2}$ decreases towards the center (since ϵ_h is decreasing with a magnetic surface radius).

The technique of calculating the bounce-averaged trapped particle drift velocity across the magnetic surfaces,[11,16] v_{an}, has been used to clarify some features of the low-collisional neoclassical transport in the original Helias. Considering the v_{an} dependence on the trapping depth we found that with decreasing trapping depth, the v_{an} value is decreasing and passes through zero at a certain trapping depth which is close to the region where the transition boundaries between the different classes of trapped particles are concentrated. This results in a decrease of the neoclassical transport coefficients in the $1/\nu$ transport regime and allows one to expect the improvement of the original Helias neoclassical properties in the low-collisional regimes where an important role belongs to the banana precession around the magnetic axis due to the radial electric field. Here the main contribution to the neoclassical transport comes from the helically trapped particle fraction in a relatively narrow region of pitch angles near the transition region between different classes of trapped particles.[2]

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